

Data Availability and Confidentiality Statement for the Alwadhi Power-Law Framework

As of January 02, 2026

The original **Alwadhi Power-Law of Diamond Pricing** (preprint available on SSRN: abstract IDs 5654570 and 5648430, posted November/October 2025) is based on a proprietary dataset of **approximately 1.2 million anonymized transaction records** spanning 2019–2024. This dataset includes:

- ~847,293 wholesale transactions from global trading centers (e.g., Antwerp, Mumbai, Tel Aviv, New York, Hong Kong, Dubai)
- ~312,457 verified online retail sales
- ~45,892 high-value auction results from major houses
- ~98,234 professional insurance appraisals

All records were contributed anonymously by industry participants (dealers, auction houses, and appraisers) under strict commercial confidentiality agreements. Sensitive information (e.g., exact certificate numbers beyond aggregates, dealer identities, or raw negotiation details) was removed during cleaning. As explicitly stated in the preprint's "Data and Code Availability" section:

"Real transaction data cannot be shared due to commercial confidentiality but summary statistics are provided for replication."

No public release, download, or de-anonymized version of this dataset exists. It remains fully proprietary and is not available on SSRN, GitHub, or any other platform.

For Replication and Extensions

- **GitHub Repository** (<https://github.com/khalilahalwadhi/The-Alwadhi-Power-Law-of-Diamond-Pricing> (<https://github.com/khalilahalwadhi/The-Alwadhi-Power-Law-of-Diamond-Pricing>)): Contains full Python implementation, example code, hard-coded parameters (e.g., $\alpha = 1.725 \pm 0.012$, shape coefficients, base price $B \approx \$3,127$), and synthetic/summary datasets that closely mimic the original's statistical properties (e.g., lognormal weight distribution, log-error $\sigma = 0.15$). Simulations provided in the repo recover the reported metrics ($R^2 \approx 0.9874$, $RMSE \approx \$498$).
- Public proxy datasets for testing/validation (much smaller but useful for power-law fits):
- Kaggle "Diamonds" datasets (~54,000 records, prices + attributes).
- Larger community variants (~220,000 graded diamonds).
- Rapaport Price List/Index (subscription benchmarks).
- PriceScope/Ritani crowdsourced trends.

Extensions to Other Gemstones (e.g., Colored Stones: Sapphires, Rubies, Emeralds)

The power-law framework is designed for modular extension to other constrained luxury commodities with scarcity-driven scaling (e.g., colored gemstones). However:

- No equivalent large-scale proprietary dataset exists yet for non-diamond gems in this work.
- Future extensions should rely on public or newly collected anonymized data (e.g., from Gemological Institute of America reports, auction aggregates like Christie's/Sotheby's colored stone sales, or platforms like GemRockAuctions).
- Recommended starting point: Adapt the model with public colored gem data (e.g., sapphire price logs vs. carat) and test for exponent convergence (hypothesized α near 1.7–2.0 depending on rarity distributions).

This statement ensures transparency for any GitHub forks, extensions (e.g., to emeralds/rubies), or related research. Contributions of anonymized colored gem data are welcome via the repo issues—contact k@maisonalwadhi.com.au for collaboration guidelines.

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Citation: Al-Wadhi, Khalilah Aisha (2025). "The Alwadhi Power-Law of Diamond Pricing: A Universal Mathematical Framework for Gemstone Valuation." SSRN Preprint.